Supplementary Material: Simulated Drives over Measured Road Profiles for Car-Based IRI Estimation

S1. Protocols, evidence classes and deviations

Two protocols. The first protocol (`evaluation/FINAL_EVALUATION_PROTOCOL.md`) and the model configurations (`config/models.json`) were frozen before any test evaluation; they define the estimators, the simulator V0, the leave-one-route-out split and the metrics (evidence class C1). After an independent review of the first version of this paper, a second protocol (`evaluation/PROTOCOL_V2.md`) was frozen before any of the new analyses was run. It lists the simulator variants V1–V10 with their seeds, the calibration and equal-budget designs, the fixed-road speed analysis, the inference rule (a paired difference is "supported" only if its 5 km-block interval excludes zero), the diagnostics and the controlled vehicle-number experiment, and it forbids choosing any variant, block size, k, draw, seed or subset after seeing its result. Four addenda (E5–E8, Section 8 of that file) were written after some of these results were known and are reported in full: the equal-budget arms for V1–V8, the paired per-draw comparison with real-only training, the speed scaling of raw vibration in simulation, and paired elasticity differences.

Evidence classes. C1: fixed before any test result. R: pre-specified re-analysis of routes whose results were already known. E: exploratory diagnostic. S: new controlled simulation with freshly generated test data. All real-data results of the main paper except the comparison of V0 with the real-trained estimators are R or E.

Deviations in the second analysis round.

- V7 and V8 enter the sensor factorial with a random half of their passes (seed 7) to match V1 and V2 in size; decided before running. Results for the full sets are in Table S12.
- The pre-specified aliasing check within one telematics unit (TMU#10) was infeasible: that unit logged at 52.6 Hz only on M13 and at 333 Hz only on CPH1 and M3. TMU#12 on M3 had no common segment $\times$ speed cell either; the check was replaced by an all-unit comparison on common cells, which is confounded with device and mounting.
- The keys (segment, task, pass) of the real units are not unique (12 duplicates where a ring-road pass covers a segment twice); earlier predictions are therefore aligned by position and checked against the labels.
- Units shorter than one Welch window have no spectrum and are excluded from the spectral cell means.

Deviations in the first analysis round.

- *Robustness experiment.* The first version drew new random passes for every perturbation level; it was re-run with common random numbers (variance reduction). The first version is kept in `results/sim/robustness_v1_independent_draws.csv`.
- *Compute allocation.* Zero-shot CNNs (three seeds, two simulation sets) and few-shot CNNs were limited before their results were known.
- *Seeds.* Neural-net metrics are those of the 5-seed ensemble prediction in the main paper; per-seed means are in `results/v2/summary_real_estimators.csv`.
- *Sensor statistics.* The telematics configuration of the simulator (55–62 Hz, 0.01 g, tilt s.d. 6°) was set from a preliminary inspection of the unlabelled LiRA-CD files. A complete re-measurement (`evaluation/lira_sensor_stats.py`) gives nominal rates of 52.6 Hz (66 % of passes) or 333 Hz, 0.01 g quantisation in 81 % of passes and tilts with median 10° (maximum 41°). V0 was not changed; the matched configurations are the variants V5 and V6.

Statements of the first version that were wrong and are corrected. (i) Real-only training was said to need "about 20 km" to match simulation; the comparator was a different simulation set, and the profile budgets were unequal (replaced by the equal-budget design). (ii) A caption called the analysis "pre-registered"; it was frozen in advance by us, not registered. (iii) "Sampling rate matters most" among sensor perturbations; the holder resonance moves the bias more. (iv) The CNN deficit was presented as general; it sits on one route. (v) Transfer was attributed to the breadth of the sensor model; the sensor factorial contradicts this (main paper, Section V-B). (vi) Urban errors were said to be smaller than the disagreement between two profilometers; that disagreement is driven by a heavy tail (Table S11), and the comparison was removed.

S2. The earlier pipeline

Labels. On the 7 LiRA-CD route-directions whose profile and IRI files are consistent, our IRI implementation reproduces the IRI reported by the P79 (an algorithm check: both are computed from the same laser profiles) with $r = 1.000$, bias $+0.018$ m/km and MAE 0.019 m/km over 1,520 segments of 100 m (Fig. S1a). For the eighth file (CPH6, direction HH) the

reported IRI cannot be aligned with the file's own profile (best $r = 0.13$ over shifts of ± 3 km, with or without reversal, against $r = 1.000$ at zero shift for two control files); all labels are therefore recomputed from the profiles. The IRI routine of the earlier pipeline is also correct (bias -0.002 m/km), but its curvature-adaptive mesh correction is not: applied to the same real profiles, which contain no mesh facets, it lowers the IRI of urban segments by 28 % on average (bias -1.10 m/km) and that of motorway segments by 6 % (Fig. S1b). On the BeamNG corpus the correction removes 64 % of the summed IRI (mean 12.50 vs 4.51 m/km) and changes the condition class of 34 % of the segments; the earlier labels were a smoothed roughness index, and the uncorrected ray-cast profiles give implausibly high values, so the corpus has no valid IRI label.

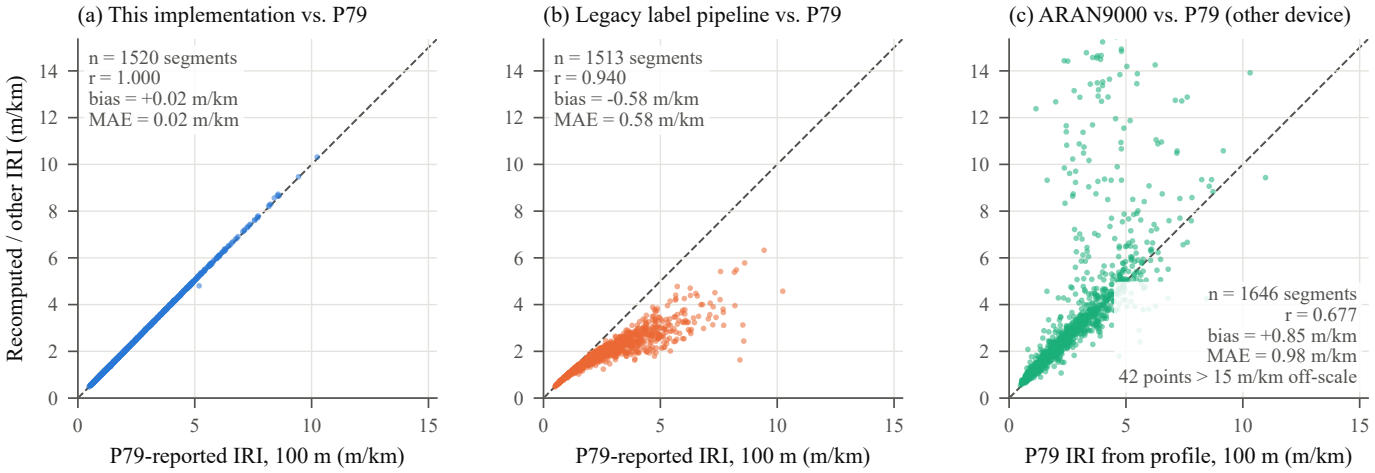


Fig. S1. Label validity on real profiles (LiRA-CD, 100 m segments, mean of both wheel paths). (a) IRI recomputed from the P79 laser profiles with our implementation vs the IRI reported by the P79. (b) The earlier label pipeline, including the curvature-adaptive mesh correction, applied to the same profiles. (c) A second profilometer (ARAN9000, other survey date) vs the P79; off-scale points are counted in the panel and included in the statistics.

Protocol. The corpus contains 19 manually driven trips of 3 BeamNG.tech vehicles on 3 maps; IMU, position and elevation were polled by unsynchronised threads, which produced exact sample-and-hold duplicates in 48–68 % of consecutive IMU rows per trip. Re-run with non-overlapping 100 m test segments and the earlier labels, the earlier recipe reaches MAE 1.67 m/km and Spearman $\rho = 0.80$ with the earlier split, in which early stopping and isotonic calibration use one trip on the test track and two further trips on the same track form the test set; under leave-one-map-out it reaches 1.99 m/km and $\rho = 0.66$, no better than an MLP on the hand-crafted features (1.90 m/km; Fig. S2). Without the calibration fitted on the test map, the exponentiated network outputs diverge on a few segments of unseen maps (MAE 251 m/km). The earlier reporting rule hid the same failure on the earlier split: discarding test predictions above 40 m/km removed 1–2 segments per seed and turned uncalibrated MAEs of up to 1,555 m/km into at most 1.94 m/km. When vehicle and map are both unseen (raw-profile labels), no estimator is substantially better than predicting the training median (MAE 7.60; vibration index 6.88; LightGBM 7.25 m/km). The earlier study's asymmetric Huber loss reduces to a scalar before Keras applies sample weights, so its "two-dimensional inverse-frequency sampling" did not re-weight individual samples; our re-implementation reproduces this behaviour.

Transfer to real cars. Trained on the BeamNG corpus and tested on LiRA-CD with the protocol of the main paper, LightGBM reaches MAE 1.36 m/km with the earlier labels and 1.97 m/km with raw-profile labels (within-route ρ 0.35 and 0.32; Table S12); its estimates rise steeply with speed on a fixed road (elasticity $+0.79$ and $+1.39$). The three-parameter vibration index trained on the same corpus ranks segments within routes far better (ρ 0.83 with the earlier labels) but misses the level (MAE 0.85 m/km, bias -0.73 m/km; Table S13), the same shape-versus-level pattern as the parametric simulator. The evaluated BeamNG pipeline did not transfer under this protocol; whether a game engine with measured road profiles and validated labels would transfer was not tested.

PVS. On the Passive Vehicular Sensors dataset (no IRI; subjective surface and quality labels), speed separates asphalt from the other surfaces with leave-one-route-out AUC 0.97 [0.94, 0.99] and "bad" from "regular" windows within cobblestone and dirt with 0.75, whereas the earlier simulation-trained model reaches 0.41 [0.33, 0.50] on the latter task (Table S2). The subjective labels are better predicted by how slowly the vehicle was driven than by how much it vibrated, so this dataset cannot validate an IRI estimator.

TABLE S1
BEAMNG CORPUS: ALL ESTIMATORS AND SPLITS (SEGMENT LEVEL, CLUSTER-BOOTSTRAP 95% CIS OVER 100 M LOCATION CELLS).

Split	Label	Model	MAE (m/km)	Spearman	R2
S3: earlier split (val. and test on test track)	legacy	M0 constant (training median)	2.68 [2.04, 3.39]	–	-0.07 [-0.19, -0.00]
S3: earlier split (val. and test on test track)	legacy	M1 speed-only (LightGBM on speed features)	2.20 [1.65, 2.85]	0.47 [0.32, 0.59]	0.14 [-0.01, 0.26]
S3: earlier split (val. and test on test track)	legacy	M2a RMS only (alpha=0)	2.25 [1.71, 2.85]	0.67 [0.55, 0.75]	0.11 [-0.25, 0.34]
S3: earlier split (val. and test on test track)	legacy	M2b RMS/v^0.5	2.10 [1.60, 2.65]	0.71 [0.60, 0.79]	0.23 [-0.04, 0.42]
S3: earlier split (val. and test on test track)	legacy	M2c RMS/v^alpha (alpha fitted on training data)	2.03 [1.56, 2.55]	0.72 [0.62, 0.79]	0.30 [0.16, 0.41]
S3: earlier split (val. and test on test track)	legacy	M2d RMS*(v_ref/v)^2 (legacy exponent, gravity removed)	2.02 [1.55, 2.53]	0.68 [0.57, 0.76]	0.29 [0.20, 0.36]
S3: earlier split (val. and test on test track)	legacy	M3 LightGBM (all features)	1.88 [1.45, 2.36]	0.74 [0.63, 0.81]	0.40 [0.18, 0.53]
S3: earlier split (val. and test on test track)	legacy	M3 LightGBM (no speed features)	1.97 [1.50, 2.48]	0.72 [0.60, 0.80]	0.35 [0.20, 0.50]
S3: earlier split (val. and test on test track)	legacy	M4 MLP	1.79 [1.34, 2.27]	0.76 [0.67, 0.83]	0.41 [0.31, 0.54]
S3: earlier split (val. and test on test track)	legacy	M5 1D-CNN	2.39 [1.47, 3.58]	0.75 [0.65, 0.82]	-3.28 [-14.77, 0.43]
S3: earlier split (val. and test on test track)	legacy	M5 1D-CNN (no speed input)	3.20 [1.69, 5.72]	0.69 [0.59, 0.77]	-17.73 [-75.41, 0.35]
S3: earlier split (val. and test on test track)	legacy	Legacy recipe (calibrated on val)	1.67 [1.29, 2.11]	0.80 [0.72, 0.85]	0.44 [0.02, 0.68]
S3: earlier split (val. and test on test track)	legacy	Legacy recipe (uncalibrated)	330.85 [1.43, 1043.50]	0.80 [0.72, 0.85]	-5902105.16 [-24835354.32, 0.63]
S1: leave-one-map-out	legacy	M0 constant (training median)	3.01 [2.61, 3.48]	-0.37 [-0.48, -0.25]	-0.21 [-0.30, -0.16]
S1: leave-one-map-out	legacy	M1 speed-only (LightGBM on speed features)	2.60 [2.26, 3.00]	0.31 [0.22, 0.39]	0.05 [-0.05, 0.13]
S1: leave-one-map-out	legacy	M2a RMS only (alpha=0)	2.33 [1.98, 2.74]	0.56 [0.51, 0.61]	0.10 [0.02, 0.18]
S1: leave-one-map-out	legacy	M2b RMS/v^0.5	2.15 [1.83, 2.52]	0.66 [0.61, 0.70]	0.22 [0.14, 0.31]
S1: leave-one-map-out	legacy	M2c RMS/v^alpha (alpha fitted on training data)	2.10 [1.79, 2.46]	0.67 [0.61, 0.71]	0.26 [0.18, 0.35]
S1: leave-one-map-out	legacy	M2d RMS*(v_ref/v)^2 (legacy exponent, gravity removed)	2.13 [1.83, 2.49]	0.64 [0.58, 0.69]	0.28 [0.20, 0.36]
S1: leave-one-map-out	legacy	M3 LightGBM (all features)	2.06 [1.77, 2.41]	0.68 [0.63, 0.72]	0.30 [0.22, 0.38]
S1: leave-one-map-out	legacy	M3 LightGBM (no speed features)	2.07 [1.77, 2.42]	0.67 [0.62, 0.71]	0.30 [0.23, 0.38]
S1: leave-one-map-out	legacy	M4 MLP	1.90 [1.64, 2.22]	0.73 [0.69, 0.77]	0.37 [0.31, 0.43]
S1: leave-one-map-out	legacy	M5 1D-CNN	2.23 [1.79, 2.77]	0.70 [0.66, 0.74]	-3.33 [-13.32, 0.35]
S1: leave-one-map-out	legacy	M5 1D-CNN (no speed input)	2.64 [1.98, 3.60]	0.62 [0.56, 0.67]	-15.92 [-60.71, 0.22]
S1: leave-one-map-out	legacy	Legacy recipe (calibrated on val)	1.99 [1.76, 2.25]	0.66 [0.59, 0.72]	0.45 [0.37, 0.50]
S1: leave-one-map-out	legacy	Legacy recipe (uncalibrated)	251.39 [1.82, 773.08]	0.68 [0.62, 0.73]	-17368673.46 [-65761393.61, 0.37]
S1: leave-one-map-out	raw profile	M0 constant (training median)	7.60 [6.67, 8.67]	-0.25 [-0.36, -0.14]	-0.15 [-0.23, -0.09]
S1: leave-one-map-out	raw profile	M1 speed-only (LightGBM on speed features)	7.49 [6.64, 8.46]	0.21 [0.12, 0.30]	-0.10 [-0.23, 0.00]
S1: leave-one-map-out	raw profile	M2a RMS only (alpha=0)	6.68 [5.90, 7.58]	0.49 [0.42, 0.55]	0.08 [-0.03, 0.17]
S1: leave-one-map-out	raw profile	M2b RMS/v^0.5	6.36 [5.62, 7.23]	0.54 [0.47, 0.60]	0.16 [0.05, 0.25]
S1: leave-one-map-out	raw profile	M2c RMS/v^alpha (alpha fitted on training data)	6.34 [5.60, 7.20]	0.54 [0.46, 0.60]	0.17 [0.06, 0.25]
S1: leave-one-map-out	raw profile	M2d RMS*(v_ref/v)^2 (legacy exponent, gravity removed)	6.50 [5.72, 7.40]	0.48 [0.38, 0.56]	0.13 [0.01, 0.22]
S1: leave-one-map-out	raw profile	M3 LightGBM (all features)	6.02 [5.37, 6.73]	0.57 [0.50, 0.63]	0.26 [0.14, 0.35]
S1: leave-one-map-out	raw profile	M3 LightGBM (no speed features)	6.01 [5.38, 6.75]	0.58 [0.51, 0.63]	0.26 [0.15, 0.35]
S1: leave-one-map-out	raw profile	M4 MLP	5.70 [5.15, 6.30]	0.59 [0.52, 0.64]	0.34 [0.19, 0.44]
S1: leave-one-map-out	raw profile	M5 1D-CNN	8.77 [5.82, 14.01]	0.51 [0.44, 0.58]	-110.29 [-387.20, 0.22]
S1: leave-one-map-out	raw profile	M5 1D-CNN (no speed input)	9.75 [5.90, 16.77]	0.52 [0.44, 0.59]	-214.49 [-752.26, 0.13]
S2: leave-one-vehicle-and-map-out	raw profile	M0 constant (training median)	7.60 [6.67, 8.67]	-0.23 [-0.34, -0.12]	-0.15 [-0.23, -0.09]
S2: leave-one-vehicle-and-map-out	raw profile	M1 speed-only (LightGBM on speed features)	7.61 [6.80, 8.55]	0.21 [0.11, 0.29]	-0.11 [-0.25, -0.00]
S2: leave-one-vehicle-and-map-out	raw profile	M2a RMS only (alpha=0)	7.27 [6.54, 8.09]	0.42 [0.36, 0.47]	-0.02 [-0.15, 0.07]
S2: leave-one-vehicle-and-map-out	raw profile	M2b RMS/v^0.5	6.88 [6.18, 7.68]	0.48 [0.41, 0.54]	0.07 [-0.07, 0.17]
S2: leave-one-vehicle-and-map-out	raw profile	M2c RMS/v^alpha (alpha fitted on training data)	6.88 [6.18, 7.69]	0.47 [0.40, 0.52]	0.07 [-0.07, 0.16]
S2: leave-one-vehicle-and-map-out	raw profile	M2d RMS*(v_ref/v)^2 (legacy exponent, gravity removed)	6.70 [5.94, 7.57]	0.45 [0.36, 0.53]	0.09 [-0.04, 0.18]
S2: leave-one-vehicle-and-map-out	raw profile	M3 LightGBM (all features)	7.25 [6.64, 7.90]	0.46 [0.39, 0.51]	0.02 [-0.17, 0.15]
S2: leave-one-vehicle-and-map-out	raw profile	M3 LightGBM (no speed features)	7.39 [6.77, 8.05]	0.45 [0.39, 0.51]	-0.01 [-0.21, 0.12]
S2: leave-one-vehicle-and-map-out	raw profile	M5 1D-CNN	11.06 [8.13, 16.02]	0.43 [0.36, 0.49]	-173.32 [-605.45, -0.78]

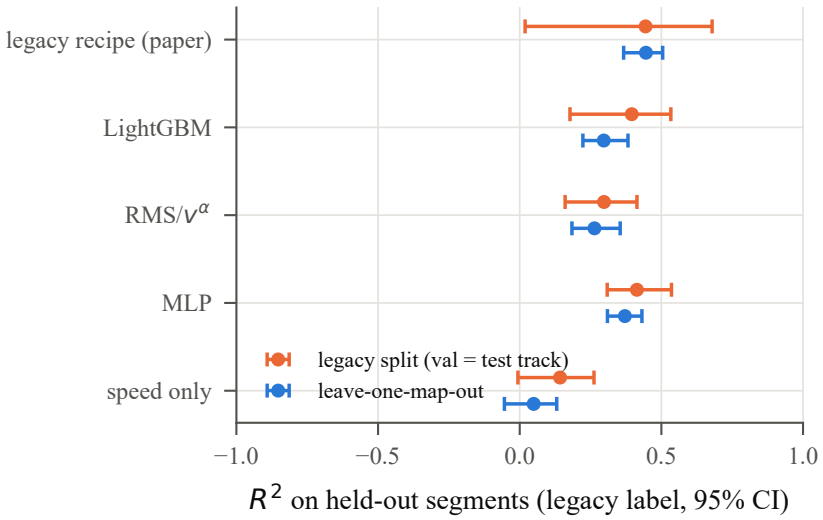


Fig. S2. BeamNG corpus, earlier label definition: R^2 on held-out segments under the earlier split (validation and test trips on the same track) and under leave-one-map-out (95 % cluster-bootstrap CIs).

TABLE S2
PVS: LEAVE-ONE-ROUTE-OUT LOGISTIC REGRESSIONS, AUC WITH CLUSTER-BOOTSTRAP 95% CIS OVER GPS CELLS.

Task	Predictors	AUC [95% CI]	windows	positives
A: asphalt vs cobblestone+dirt	speed	0.97 [0.94, 0.99]	1070	352
A: asphalt vs cobblestone+dirt	RMS	0.93 [0.91, 0.94]	1070	352
A: asphalt vs cobblestone+dirt	RMS/sqrt(v)	0.99 [0.98, 1.00]	1070	352
A: asphalt vs cobblestone+dirt	model	0.94 [0.90, 0.96]	1070	352
A: asphalt vs cobblestone+dirt	model+speed	0.99 [0.97, 1.00]	1070	352
A: asphalt vs cobblestone+dirt	model+RMS	0.96 [0.95, 0.98]	1070	352
A: asphalt vs cobblestone+dirt	model+speed+RMS	0.99 [0.99, 1.00]	1070	352
A: asphalt vs cobblestone+dirt	speed+RMS	0.99 [0.98, 1.00]	1070	352
B: good vs regular+bad	speed	0.97 [0.95, 0.99]	1070	358
B: good vs regular+bad	RMS	0.93 [0.91, 0.95]	1070	358
B: good vs regular+bad	RMS/sqrt(v)	0.99 [0.98, 1.00]	1070	358
B: good vs regular+bad	model	0.94 [0.90, 0.96]	1070	358
B: good vs regular+bad	model+speed	0.99 [0.97, 1.00]	1070	358
B: good vs regular+bad	model+RMS	0.97 [0.95, 0.98]	1070	358
B: good vs regular+bad	model+speed+RMS	1.00 [0.99, 1.00]	1070	358
B: good vs regular+bad	speed+RMS	0.99 [0.98, 1.00]	1070	358
C: bad vs regular (cobblestone+dirt only)	speed	0.75 [0.67, 0.82]	352	72
C: bad vs regular (cobblestone+dirt only)	RMS	0.23 [0.15, 0.32]	352	72
C: bad vs regular (cobblestone+dirt only)	RMS/sqrt(v)	0.43 [0.35, 0.51]	352	72
C: bad vs regular (cobblestone+dirt only)	model	0.41 [0.33, 0.50]	352	72
C: bad vs regular (cobblestone+dirt only)	model+speed	0.75 [0.67, 0.82]	352	72
C: bad vs regular (cobblestone+dirt only)	model+RMS	0.51 [0.42, 0.58]	352	72
C: bad vs regular (cobblestone+dirt only)	model+speed+RMS	0.80 [0.72, 0.86]	352	72
C: bad vs regular (cobblestone+dirt only)	speed+RMS	0.78 [0.70, 0.84]	352	72

S3. SIMULATOR

The half-car is discretised exactly under a first-order hold and solved in modal coordinates; the solution deviates by at most $3.9 \cdot 10^{-13}$ (relative) from a direct state-space recursion and by $4.4 \cdot 10^{-9}$ from a fine-step Runge–Kutta integration. The sampled vehicles have body modes at 0.95–2.52 Hz and wheel-hop modes at 9.1–16.1 Hz. Speed is drawn independently of roughness (base speed 20–110 km/h with Ornstein–Uhlenbeck fluctuations and random braking events). Table S3 lists the parameter ranges; the variants and their seeds are listed in `evaluation/PROTOCOL_V2.md` and generated by `experiments/sim_generate.py`.

TABLE S3
PARAMETER RANGES OF THE PARAMETRIC SIMULATOR (SENSITIVITY RANGES; SOURCES IN CONFIG/SIM_RANGES.JSON).

group	parameter	range	unit	source
vehicle	m_s	1000 - 1800	kg	G, Z
vehicle	r_y	1 - 1.3	m	G (pitch radius of gyration ~ sqrt(a*b))
vehicle	L	2.4 - 3	m	G, Z
vehicle	a_frac	0.4 - 0.5	-	front-heavy passenger cars
vehicle	f_s	1 - 1.6	Hz	G, S
vehicle	rear_freq_ratio	1 - 1.2	-	G (flat-ride tuning)
vehicle	zeta	0.2 - 0.45	-	G, S
vehicle	m_u_per_wheel	30 - 50	kg	G
vehicle	k_t_per_tire	150000 - 300000	N/m	G (wheel hop 10-15 Hz)
vehicle	x_s	-0.5 - 1.2	m	sensor between CG and front axle (LiRA readme: near front axle)
sensor: autopi	rate / LSB / noise / tilt SD	55-62 Hz / 0.01 g / 150-400 ug/rtHz / 6.0 deg	-	M, D
sensor: phone	rate / LSB / noise / tilt SD	90-210 Hz / 0.0006 g / 100-300 ug/rtHz / 12.0 deg	-	D (smartphone IMUs); cradle resonance range is an assumption, treated as sensitivity
speed	base speed	20-110	km/h	design choice (independent of roughness)

S4. REAL CARS: ADDITIONAL RESULTS

Table S4 gives every estimator per held-out route. Table S5 gives paired differences at four levels of dependence: segment, 1 km and 5 km blocks, and routes; Fig. S3a shows why segment-level intervals are too narrow. Table S6 decomposes pass-level errors into segment, telematics-unit and residual components. Table S7 gives the unit-disjoint baseline, Table S8 the within-road anchor to Bajic et al., Table S9 the effect of averaging passes on a fixed set of segments, Table S10 the calibrated quarter-car inversion per fold (with the fitted transmissibility in Fig. S3b) and Table S11 the comparison of the two profilometers.

TABLE S4
PER-ROUTE RESULTS ON REAL CARS (SEGMENT MAE IN M/KM AND WITHIN-ROUTE SPEARMAN ρ; EACH HELD-OUT ROUTE EVALUATED BY THE MODELS TRAINED ON THE OTHER THREE; ρ IS UNDEFINED FOR PREDICTORS THAT ARE CONSTANT WITHIN A ROUTE). SEGMENTS: CPH1 509, CPH6 426, M3 515, M13 228 (M7: SEGMENTS WITH A USABLE ROAD SPECTRUM). LAST ROWS: UNTRAINED VIBRATION STATISTICS, WHOSE WITHIN-ROUTE ρ NEEDS NO TRAINING DATA.

	MAE				ρ			
Estimator	CPH1	CPH6	M3	M13	CPH1	CPH6	M3	M13
M0 constant	2.12	2.03	1.31	0.79	–	–	–	–
M1 speed only	1.10	1.11	0.31	0.25	0.48	0.42	0.20	0.45
M2c vibration index	1.03	0.86	0.35	0.25	0.74	0.74	0.86	0.70
M3 LightGBM	0.60	0.70	0.13	0.15	0.88	0.84	0.90	0.72
M3 LightGBM, no speed	1.03	0.74	0.35	0.35	0.80	0.80	0.86	0.57
M4 MLP	0.60	0.70	0.13	0.16	0.88	0.83	0.89	0.72
M5 1D-CNN	0.73	0.68	0.15	0.17	0.87	0.79	0.88	0.71
Sim V0 LightGBM (zero-shot)	0.55	0.73	0.21	0.18	0.88	0.81	0.93	0.73
Sim V10 LightGBM (zero-shot)	0.57	0.69	0.26	0.21	0.88	0.81	0.93	0.73
M0b map-only road-class prior	1.20	1.18	0.34	0.30	–	–	–	–
M7 quarter-car inversion	0.69	0.71	0.27	0.17	0.86	0.82	0.89	0.75
M7 + output calibration	0.70	0.87	0.47	0.23	0.86	0.82	0.89	0.75
untrained: raw RMS	–	–	–	–	0.71	0.65	0.90	0.75
untrained: RMS/$v^{0.5}$	–	–	–	–	0.84	0.79	0.92	0.74
untrained: RMS/v	–	–	–	–	0.86	0.78	0.92	0.73

TABLE S5
PAIRED MAE DIFFERENCES (M/KM) AT FOUR LEVELS OF DEPENDENCE: CLUSTER BOOTSTRAP OVER SEGMENTS, 1 KM AND 5 KM BLOCKS (B = 2000), AND THE ROUTE-LEVEL DESCRIPTIVE INTERVAL (MEAN OF THE 4 PER-ROUTE DIFFERENCES, t WITH 3 DF). REFERENCE: REAL-TRAINED LIGHTGBM UNLESS STATED.

Estimator	Diff.	Segment	1 km block	5 km block	Route (t_3)
V0 LightGBM (zero-shot)	+0.019	[+0.008, +0.031]	[+0.002, +0.036]	[−0.003, +0.040]	[−0.06, +0.10]
V10 LightGBM (zero-shot)	+0.039	[+0.026, +0.052]	[+0.019, +0.059]	[+0.007, +0.068]	[−0.07, +0.15]
MLP (per-seed mean)	+0.005	[−0.003, +0.012]	[−0.007, +0.017]	[−0.012, +0.024]	[+0.00, +0.01]
MLP (ensemble)	+0.004	[−0.004, +0.011]	[−0.008, +0.016]	[−0.013, +0.023]	[+0.00, +0.01]
1D-CNN (per-seed mean)	+0.055	[+0.042, +0.068]	[+0.035, +0.076]	[+0.029, +0.086]	[−0.07, +0.16]
1D-CNN (ensemble)	+0.045	[+0.032, +0.059]	[+0.025, +0.066]	[+0.019, +0.075]	[−0.07, +0.15]
speed only	+0.324	[+0.295, +0.351]	[+0.277, +0.373]	[+0.252, +0.404]	[−0.01, +0.60]
vibration index M2c	+0.251	[+0.226, +0.275]	[+0.212, +0.292]	[+0.202, +0.306]	[−0.00, +0.46]
road-class prior (map)	+0.387	[+0.356, +0.418]	[+0.339, +0.438]	[+0.307, +0.475]	[+0.02, +0.70]
road-class prior (map) vs speed only	+0.064	[+0.045, +0.082]	[+0.028, +0.100]	[+0.029, +0.101]	[+0.02, +0.11]

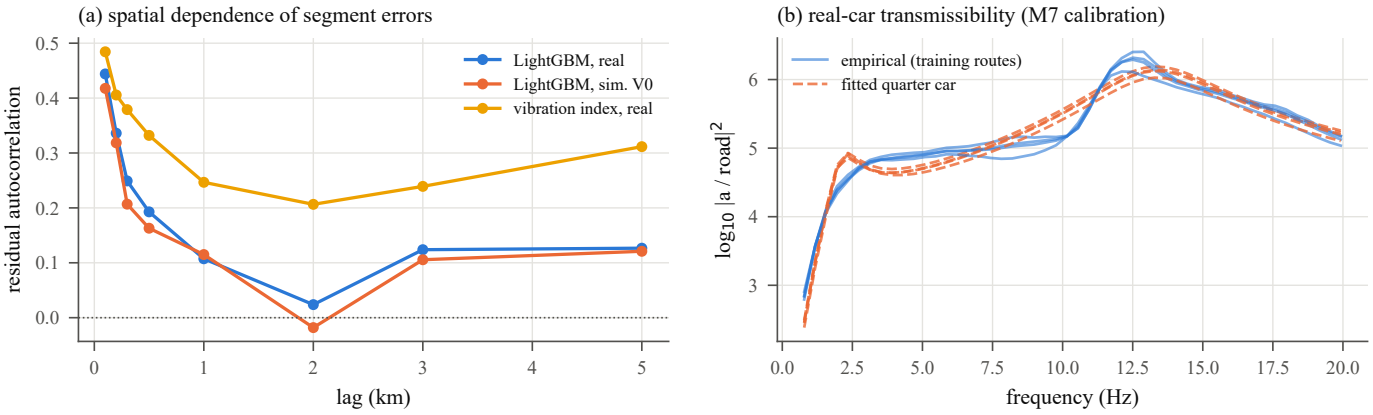


Fig. S3. (a) Autocorrelation of segment-level log errors along the road, by lag. (b) Empirical transmissibility of the real car (acceleration spectrum divided by the road displacement spectrum at the pass speed; training routes of each fold) and the fitted quarter car used by M7.

TABLE S6

VARIANCE COMPONENTS OF PASS-LEVEL LOG ERRORS (TWO-WAY DECOMPOSITION: SEGMENT EFFECTS + TELEMATICS-UNIT EFFECTS; SD IN % OF LOG ERROR; UNIT SD WITH A 1 KM-BLOCK BOOTSTRAP INTERVAL).

Model	Unit (vehicle) SD	Segment SD	Pass residual SD	Units
REAL GBM	7.3 [6.6, 8.5]	21.7	15.2	9
SIM P79	7.7 [6.8, 9.6]	21.8	16.2	9

TABLE S7

UNIT-DISJOINT REAL BASELINE: LIGHTGBM TRAINED ONLY ON PASSES OF TELEMATICS UNITS THAT NEVER DRIVE THE HELD-OUT ROUTE, VS A SIZE-MATCHED RANDOM SUBSET OF ALL UNITS (5 DRAWS; MEAN [MIN, MAX]) AND THE FULL V1 MODEL. THE M3 FOLD KEEPS ONLY URBAN TRAINING DATA AND IS NOT INTERPRETABLE AS A VEHICLE EFFECT.

Held-out route	Train units	Classes	Unit-disjoint	Size-matched	All units (v1)
CPH1	3,176	motorway, urban	0.63	0.63 [0.60, 0.65]	0.60
CPH6	11,976	motorway, urban	0.69	0.70 [0.69, 0.71]	0.70
M3	2,924	urban	0.62	0.16 [0.14, 0.19]	0.13
M13	7,323	motorway, urban	0.15	0.15 [0.15, 0.15]	0.15

TABLE S8

APPROXIMATE ANCHOR TO BAJIC ET AL. (2021): WITHIN-ROUTE SPATIAL HOLDOUT ON M3 (EACH DIRECTION: FIRST 80 % OF THE CHAINAGE FOR TRAINING, LAST 20 % FOR TESTING). BAJIC ET AL. REPORTED MAE 0.16–0.17, RMSE 0.21 AND R^2 0.64–0.66 FOR THEIR BEST MODELS (AND MAE 0.33 FOR THE MEAN BASELINE) WITH A DIFFERENT PROFILER, OVERLAPPING WINDOWS AND THEIR OWN PASS SELECTION, SO THE NUMBERS ARE NOT DIRECTLY COMPARABLE.

Model	MAE	RMSE	R^2
train mean (baseline)	0.319	0.391	−0.02
LightGBM	0.115	0.171	0.81
vibration index (alpha fitted)	0.133	0.185	0.77
MLP (5-seed mean prediction)	0.119	0.175	0.80

TABLE S9

SEGMENT MAE VS NUMBER OF PASSES AVERAGED PER SEGMENT, ON THE FIXED SET OF SEGMENTS WITH ≥8 PASSES (20 RANDOM PASS SUBSETS).

Model	Road	1	2	3	5	8
real LightGBM	motorway	0.20	0.17	0.16	0.15	0.14
real LightGBM	urban	0.66	0.60	0.59	0.57	0.56
real index (M2c)	motorway	0.33	0.33	0.32	0.32	0.32
real index (M2c)	urban	1.19	1.10	1.06	1.03	1.00
sim LightGBM (V0)	motorway	0.24	0.22	0.21	0.20	0.20
sim LightGBM (V0)	urban	0.64	0.56	0.54	0.53	0.52

TABLE S10
CALIBRATED QUARTER-CAR SPECTRAL INVERSION (M7) PER HELD-OUT ROUTE: FITTED NORMALISED QUARTER CAR (FIT TO THE EMPIRICAL TRANSMISSIBILITY OF THE THREE TRAINING ROUTES) AND SEGMENT MAE WITHOUT AND WITH THE LOG-LINEAR OUTPUT CALIBRATION.

Route	K_1	K_2	C	U	gain	fit RMSE	MAE	MAE, calibrated
CPH1	1932	191	7.3	0.29	0.63	0.42	0.69	0.70
CPH6	1911	186	7.3	0.29	0.60	0.41	0.71	0.87
M3	1791	202	6.9	0.28	0.56	0.42	0.27	0.47
M13	1876	189	7.2	0.29	0.60	0.41	0.17	0.23

TABLE S11
ARAN9000 VS P79 (100 M, MEAN OF BOTH WHEEL PATHS) AFTER HEADING-CONSTRAINED MATCHING AND REMOVAL OF ZERO-IRI RECORDS. SURVEYS: P79 10-09-2020; ARAN 12-08-2020 (CPH1, M3, M13) AND 30-08-2021 (CPH6).

Scope	Segments	MAE	Bias	Pearson	Spearman	Median ratio	>15 m/km (%)
all	1645	0.98	+0.85	0.68	0.96	1.10	2.7
CPH1	483	1.51	+1.30	0.60	0.87	1.08	3.9
CPH6	410	1.89	+1.63	0.50	0.80	1.06	6.1
M13	230	0.12	+0.09	0.96	0.93	1.10	0.0
M3	522	0.15	+0.14	0.95	0.94	1.12	0.0
urban	893	1.68	+1.45	0.54	0.84	1.07	4.9
motorway	752	0.14	+0.12	0.95	0.95	1.12	0.0

S5. SIMULATOR VARIANTS: ADDITIONAL RESULTS

Tables S12 and S13 list all variants for LightGBM and for the vibration index trained in simulation, including the full V7 and V8 sets and the BeamNG corpus. Fig. S4 shows segment-level predictions of the real-trained model and of three variants. Table S14 compares every measured-profile variant with real-only training on the same k km, draw by draw.

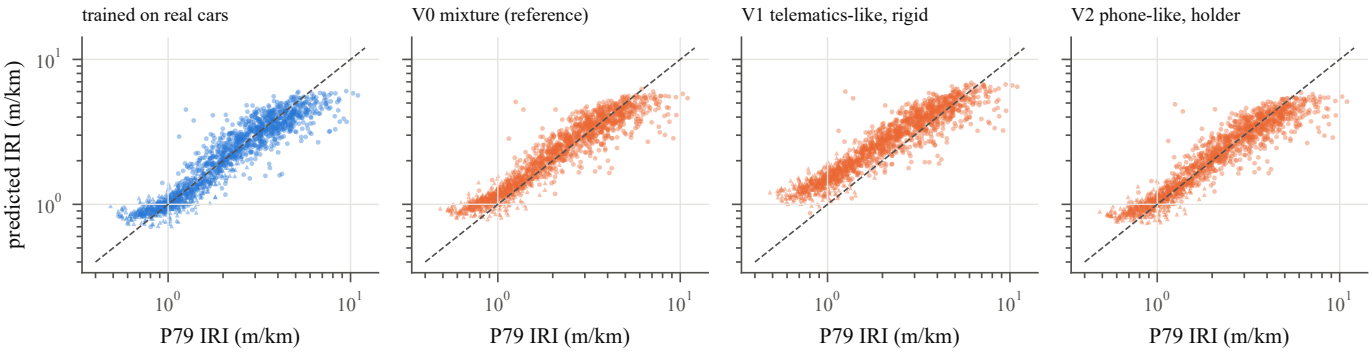


Fig. S4. Predicted vs P79 IRI on the held-out routes (segment means; circles urban, triangles motorway; log axes): LightGBM trained on real cars and trained only on simulator variants V0, V1 and V2.

TABLE S12

ALL SIMULATOR VARIANTS (LIGHTGBM), INCLUDING THE FULL V7/V8 SETS (ALL PASSES INSTEAD OF A RANDOM HALF) AND THE BEAMNG CORPUS WITH THE EARLIER (CORRECTED) AND RAW-PROFILE LABELS. COLUMNS AS IN TABLE II: ZERO-SHOT MAE AND BIAS, WITHIN-ROUTE SPEARMAN, SCALE b, POOLED WITHIN-SEGMENT SPEED ELASTICITY, AND MAE AFTER A LEVEL OFFSET FROM k KM OF REAL DATA OF THE TRAINING ROUTES.

Simulator variant	Zero-shot	Bias (m/km)		Within-	Scale	Speed	+ level from k km real		
	MAE	urban	motorway	route ρ	b	elasticity	1 km	5 km	20 km
V0 mixture (reference, pre-specified)	0.44	−0.12	+0.17	0.83	0.69	−0.06	0.49	0.42 [0.40, 0.45]	0.42
V1 telematics-like, rigid	0.68	+0.32	+0.56	0.83	0.66	+0.02	0.59	0.53 [0.47, 0.60]	0.54
V7 telematics-like + holder	0.46	−0.05	+0.23	0.83	0.70	−0.15	0.50	0.43 [0.40, 0.46]	0.43
V8 phone-like, rigid	0.55	+0.21	+0.34	0.84	0.71	−0.00	0.51	0.45 [0.42, 0.49]	0.46
V2 phone-like + holder	0.41	−0.22	+0.06	0.83	0.69	−0.10	0.48	0.42 [0.40, 0.45]	0.41
V3 ideal sensor	0.67	+0.27	+0.56	0.83	0.62	+0.01	0.60	0.54 [0.48, 0.62]	0.55
V5 telematics matched to LiRA-CD	0.69	+0.25	+0.60	0.82	0.64	+0.01	0.62	0.56 [0.51, 0.64]	0.58
V6 matched telematics + phone-like	0.46	−0.24	+0.22	0.83	0.69	+0.06	0.52	0.45 [0.41, 0.48]	0.45
V4 one mid-range vehicle	0.63	+0.39	+0.41	0.83	0.66	−0.05	0.50	0.44 [0.41, 0.46]	0.43
V10 measured + synthetic profiles	0.46	−0.26	+0.23	0.84	0.66	−0.03	0.53	0.46 [0.41, 0.50]	0.46
V9 synthetic profiles only	0.71	+0.09	+0.70	0.81	0.66	+0.23	0.84	0.75 [0.65, 0.88]	0.77
V7, all passes	0.50	+0.00	+0.30	0.84	0.68	−0.11	0.52	0.45 [0.41, 0.49]	0.46
V8, all passes	0.55	+0.19	+0.32	0.83	0.70	+0.02	0.51	0.45 [0.42, 0.49]	0.46
BeamNG corpus, earlier labels	1.36	−2.21	−0.08	0.35	0.48	+0.79	1.77	1.59 [1.30, 1.87]	1.66
BeamNG corpus, raw-profile labels	1.97	−0.47	+2.94	0.32	0.42	+1.39	2.56	2.22 [1.66, 2.85]	2.42
trained on real cars (M3)	0.42	−0.19	−0.02	0.83	0.66	−0.15	–	–	–

TABLE S13

ALL SIMULATOR VARIANTS (VIBRATION INDEX WITH THE SPEED EXPONENT FITTED IN SIMULATION), INCLUDING THE FULL V7/V8 SETS (ALL PASSES INSTEAD OF A RANDOM HALF) AND THE BEAMNG CORPUS WITH THE EARLIER (CORRECTED) AND RAW-PROFILE LABELS. COLUMNS AS IN TABLE II: ZERO-SHOT MAE AND BIAS, WITHIN-ROUTE SPEARMAN, SCALE b, POOLED WITHIN-SEGMENT SPEED ELASTICITY, AND MAE AFTER A LEVEL OFFSET FROM k KM OF REAL DATA OF THE TRAINING ROUTES.

Simulator variant	Zero-shot	Bias (m/km)		Within-	Scale	Speed	+ level from k km real		
	MAE	urban	motorway	route ρ	b	elasticity	1 km	5 km	20 km
V0 mixture (reference, pre-specified)	0.57	+0.24	+0.28	0.81	0.75	−0.48	0.57	0.52 [0.48, 0.55]	0.52
V1 telematics-like, rigid	0.66	+0.53	+0.36	0.81	0.76	−0.49	0.56	0.51 [0.47, 0.54]	0.51
V7 telematics-like + holder	0.56	+0.24	+0.28	0.82	0.75	−0.45	0.56	0.51 [0.47, 0.55]	0.51
V8 phone-like, rigid	0.63	+0.44	+0.34	0.81	0.76	−0.47	0.58	0.52 [0.48, 0.56]	0.52
V2 phone-like + holder	0.55	+0.13	+0.21	0.81	0.77	−0.49	0.58	0.52 [0.49, 0.56]	0.53
V3 ideal sensor	0.63	+0.38	+0.40	0.81	0.72	−0.45	0.58	0.52 [0.48, 0.57]	0.53
V5 telematics matched to LiRA-CD	0.63	+0.33	+0.42	0.82	0.73	−0.42	0.58	0.52 [0.47, 0.57]	0.53
V6 matched telematics + phone-like	0.54	+0.05	+0.33	0.82	0.72	−0.40	0.59	0.53 [0.48, 0.58]	0.53
V4 one mid-range vehicle	0.74	+0.68	+0.42	0.82	0.78	−0.45	0.57	0.51 [0.47, 0.55]	0.52
V10 measured + synthetic profiles	0.57	+0.18	+0.36	0.82	0.74	−0.39	0.59	0.53 [0.47, 0.58]	0.53
V9 synthetic profiles only	0.69	−0.20	+0.74	0.83	0.65	+0.05	0.98	0.92 [0.78, 1.07]	0.95
V7, all passes	0.55	+0.12	+0.31	0.82	0.73	−0.41	0.58	0.53 [0.48, 0.58]	0.53
V8, all passes	0.63	+0.44	+0.33	0.81	0.77	−0.47	0.58	0.52 [0.48, 0.55]	0.52
BeamNG corpus, earlier labels	0.85	−1.37	+0.08	0.83	0.50	−0.11	1.04	0.97 [0.85, 1.14]	1.00
BeamNG corpus, raw-profile labels	1.96	+1.51	+2.36	0.82	0.46	+0.08	1.25	1.18 [1.02, 1.40]	1.23
trained on real cars (M3)	0.67	−0.44	+0.26	0.76	0.50	−0.64	–	–	–

TABLE S14

EVERY MEASURED-PROFILE SIMULATOR VARIANT AGAINST LIGHTGBM TRAINED ON THE REAL PASSES OVER THE SAME k KM (20 PAIRED DRAWS OF TRAINING-ROUTE BLOCKS). ENTRIES: MEAN DIFFERENCE IN POOLED SEGMENT MAE, SIMULATION ARM MINUS REAL-ONLY (M/KM; NEGATIVE = SIMULATION BETTER), AND IN PARENTHESES THE NUMBER OF THE 20 DRAWS IN WHICH THE SIMULATION ARM WAS BETTER. "SAME k KM": TRAINED ON SIMULATED PASSES OVER THE PROFILES OF THE SAME BLOCKS ONLY (EQUAL PROFILE BUDGET; V1–V8 ADDED AFTER THE V0 RESULT WAS KNOWN, CLASS E). "ALL PROFILES + LEVEL": TRAINED ON ALL TRAINING-ROUTE PROFILES, LEVEL OFFSET FROM THE k KM OF REAL PASSES. "+ AFFINE": AFFINE CALIBRATION INSTEAD OF THE OFFSET.

	Same k km of profiles			All profiles + level from k km				+ affine	
Variant	2 km	5 km	20 km	1 km	5 km	10 km	20 km	5 km	20 km
V0	−0.28 (20)	−0.08 (16)	+0.01 (9)	−1.14 (20)	−0.25 (20)	−0.10 (20)	−0.04 (19)	−0.20 (19)	−0.03 (17)
V1	−0.09 (16)	+0.11 (2)	+0.24 (0)	−1.04 (20)	−0.15 (14)	+0.02 (5)	+0.08 (0)	−0.18 (18)	+0.02 (7)
V2	−0.27 (20)	−0.08 (16)	−0.02 (17)	−1.15 (20)	−0.26 (20)	−0.11 (20)	−0.05 (20)	−0.21 (19)	−0.03 (19)
V3	−0.16 (18)	+0.06 (4)	+0.22 (0)	−1.03 (20)	−0.14 (14)	+0.04 (4)	+0.09 (0)	−0.20 (19)	−0.01 (12)
V4	−0.17 (18)	+0.03 (6)	+0.19 (0)	−1.13 (20)	−0.24 (20)	−0.09 (20)	−0.03 (18)	−0.21 (20)	−0.02 (17)
V5	−0.12 (16)	+0.09 (3)	+0.23 (0)	−1.01 (20)	−0.12 (13)	+0.06 (3)	+0.12 (0)	−0.18 (18)	+0.02 (5)
V6	−0.24 (20)	−0.05 (13)	+0.02 (4)	−1.11 (20)	−0.23 (20)	−0.07 (17)	−0.01 (14)	−0.21 (19)	−0.02 (16)
V7	−0.22 (19)	−0.03 (10)	+0.05 (0)	−1.13 (20)	−0.25 (20)	−0.09 (20)	−0.03 (18)	−0.20 (18)	−0.01 (13)
V8	−0.21 (18)	−0.01 (9)	+0.10 (0)	−1.12 (20)	−0.23 (20)	−0.06 (18)	−0.00 (14)	−0.20 (19)	−0.02 (14)
Real-only MAE	1.14	0.68	0.46	1.63	0.68	0.52	0.46	0.68	0.46

S6. Spectra and speed: additional results

Table S15 gives the band means of the spectral differences in Fig. 3a, Table S16 the 10–15 Hz ratio by road class and the speed scaling of raw vibration in simulation and reality, and Table S17 the fixed-road speed elasticities of all estimators with paired differences to the real-trained estimators. In the 333 Hz passes, 25 % of the vibration energy above 0.3 Hz lies at 10–20 Hz, 21 % above 20 Hz and 13 % above the 26.3 Hz Nyquist frequency of the slower logging (medians over 5,721 units).

TABLE S15

Mean $\log_{10}$ PSD difference by band, real minus simulated (re-simulated passes of the first 10 training vehicles over all P79 profiles), on common segment × 10 km/h cells; last row: real passes logged at 52.6 Hz minus 333 Hz (all units, device-confounded).

Comparison	0.5–2 Hz	2–5 Hz	5–10 Hz	10–15 Hz	15–20 Hz	Cells
all units: 52.6Hz-333Hz	−0.06	−0.09	−0.16	−0.24	−0.40	1,765
real-V1	−0.18	+0.38	+0.01	+0.69	+0.89	3,577
real-V2	−0.20	+0.39	−0.14	+0.27	+0.06	3,716
real-V3	−0.19	+0.49	+0.08	+0.71	+0.97	3,455
real-V5	−0.20	+0.55	+0.12	+0.69	+0.91	3,127
real-V7	−0.18	+0.43	−0.01	+0.47	+0.48	3,574
real-V8	−0.15	+0.37	−0.05	+0.45	+0.48	3,277

TABLE S16

Simulated vs real vibration on the same segments, per variant. Left: real/simulated amplitude ratio of the 10–15 Hz band (cell means over segment × 10 km/h speed band; all routes, urban, motorway) next to the zero-shot bias. Right: speed scaling of the raw vibration RMS on a fixed road (within-segment elasticity; simulated passes restricted to the speed range of the real passes over the same segment; 95 % CI from a 5 km-block bootstrap).

	10–15 Hz ratio real/sim			Zero-shot	RMS speed elasticity	
Source	all	urban	motorway	bias	urban	motorway
V0	1.86	1.57	2.14	+0.01	1.25 [1.17, 1.34]	1.10 [1.02, 1.18]
V1	2.41	1.98	2.81	+0.43	1.22 [1.13, 1.31]	1.07 [1.01, 1.17]
V2	1.36	1.33	1.39	−0.09	1.30 [1.19, 1.42]	1.29 [1.20, 1.34]
V3	2.37	1.92	2.81	+0.40	1.16 [1.10, 1.23]	1.40 [1.31, 1.46]
V4	1.93	1.67	2.19	+0.40	1.11 [1.07, 1.15]	1.26 [1.12, 1.41]
V5	2.26	1.93	2.60	+0.40	0.92 [0.84, 1.01]	1.32 [1.21, 1.41]
V6	1.80	1.49	2.12	−0.04	0.95 [0.85, 1.06]	1.17 [1.07, 1.25]
V7	1.80	1.57	2.00	+0.07	1.27 [1.14, 1.39]	1.27 [1.12, 1.35]
V8	1.89	1.60	2.15	+0.27	1.05 [0.97, 1.10]	1.21 [1.08, 1.31]
Real car	1	1	1	–	0.60 [0.50, 0.69]	0.85 [0.74, 0.93]

TABLE S17

SPEED ELASTICITY OF EACH ESTIMATE ON A FIXED ROAD: SLOPE OF LOG(PREDICTION) ON LOG(SPEED) WITH SEGMENT FIXED EFFECTS (SEGMENTS WITH $\geq$3 PASSES), 95 % CI BY 1 KM-BLOCK BOOTSTRAP; "BETWEEN": DESIGN WITH ROUTE EFFECTS AND LOG IRI AS COVARIATE. |ELASTICITY| $\leq$0.104 CORRESPONDS TO THE $\pm$10 % CRITERION OVER 40–100 KM/H. LAST COLUMN: PAIRED DIFFERENCE TO THE REAL-TRAINED ESTIMATOR OF THE SAME FAMILY (LIGHTGBM OR INDEX) ON THE SAME PASSES, 95 % CI BY 5 KM-BLOCK BOOTSTRAP (POOLED).

Estimate	Pooled	Urban	Motorway	Between	− real-trained
RAW RMS	+0.66 [+0.60, +0.71]	+0.59 [+0.53, +0.65]	+0.84 [+0.78, +0.91]	+0.64	–
REAL SPEED	−0.46 [−0.55, −0.37]	−0.25 [−0.29, −0.21]	−0.99 [−1.12, −0.88]	−0.46	–
REAL INDEX	−0.64 [−0.68, −0.59]	−0.68 [−0.73, −0.63]	−0.51 [−0.54, −0.49]	−0.65	–
REAL RMS index	+0.21 [+0.18, +0.24]	+0.20 [+0.16, +0.24]	+0.26 [+0.24, +0.28]	+0.21	–
REAL GBM	−0.15 [−0.19, −0.11]	−0.10 [−0.14, −0.07]	−0.29 [−0.35, −0.21]	−0.16	–
REAL GBM nospeed	−0.14 [−0.18, −0.10]	−0.21 [−0.24, −0.18]	+0.02 [−0.07, +0.18]	−0.17	–
REAL MLP	−0.19 [−0.23, −0.15]	−0.14 [−0.19, −0.10]	−0.33 [−0.38, −0.27]	−0.19	–
REAL CNN	−0.19 [−0.24, −0.15]	−0.12 [−0.16, −0.09]	−0.37 [−0.42, −0.31]	−0.19	–
SIM V0 GBM	−0.06 [−0.11, −0.01]	+0.02 [−0.03, +0.06]	−0.27 [−0.32, −0.21]	−0.08	+0.09 [+0.06, +0.12]
SIM V0 INDEX	−0.48 [−0.53, −0.44]	−0.53 [−0.58, −0.47]	−0.37 [−0.42, −0.30]	−0.50	+0.15 [+0.12, +0.18]
SIM V1 GBM	+0.02 [−0.03, +0.07]	+0.09 [+0.04, +0.14]	−0.17 [−0.26, −0.10]	−0.01	+0.17 [+0.13, +0.20]
SIM V1 INDEX	−0.49 [−0.54, −0.44]	−0.54 [−0.59, −0.48]	−0.36 [−0.42, −0.30]	−0.51	+0.15 [+0.11, +0.17]
SIM V2 GBM	−0.10 [−0.16, −0.05]	−0.01 [−0.05, +0.03]	−0.36 [−0.41, −0.30]	−0.11	+0.05 [+0.01, +0.08]
SIM V2 INDEX	−0.49 [−0.55, −0.44]	−0.55 [−0.61, −0.49]	−0.35 [−0.40, −0.28]	−0.52	+0.14 [+0.11, +0.17]
SIM V3 GBM	+0.01 [−0.04, +0.05]	+0.09 [+0.05, +0.13]	−0.22 [−0.29, −0.16]	−0.02	+0.16 [+0.12, +0.19]
SIM V3 INDEX	−0.45 [−0.50, −0.40]	−0.51 [−0.56, −0.45]	−0.30 [−0.36, −0.24]	−0.48	+0.19 [+0.15, +0.21]
SIM V4 GBM	−0.05 [−0.10, −0.00]	+0.03 [−0.01, +0.07]	−0.26 [−0.32, −0.21]	−0.05	+0.10 [+0.08, +0.13]
SIM V4 INDEX	−0.45 [−0.50, −0.40]	−0.50 [−0.56, −0.44]	−0.30 [−0.36, −0.24]	−0.47	+0.19 [+0.14, +0.22]
SIM V5 GBM	+0.01 [−0.03, +0.05]	+0.08 [+0.04, +0.12]	−0.16 [−0.25, −0.10]	−0.01	+0.17 [+0.13, +0.20]
SIM V5 INDEX	−0.42 [−0.47, −0.37]	−0.48 [−0.54, −0.43]	−0.24 [−0.30, −0.18]	−0.44	+0.22 [+0.19, +0.25]
SIM V6 GBM	+0.06 [+0.01, +0.11]	+0.16 [+0.11, +0.20]	−0.19 [−0.27, −0.12]	+0.05	+0.21 [+0.18, +0.24]
SIM V6 INDEX	−0.40 [−0.45, −0.35]	−0.45 [−0.50, −0.39]	−0.26 [−0.31, −0.19]	−0.42	+0.24 [+0.21, +0.26]
SIM V7 GBM	−0.15 [−0.20, −0.11]	−0.10 [−0.16, −0.06]	−0.28 [−0.33, −0.22]	−0.18	+0.00 [−0.04, +0.03]
SIM V7 INDEX	−0.45 [−0.50, −0.40]	−0.50 [−0.56, −0.44]	−0.33 [−0.39, −0.26]	−0.47	+0.18 [+0.15, +0.21]
SIM V8 GBM	−0.00 [−0.05, +0.05]	+0.08 [+0.03, +0.13]	−0.22 [−0.28, −0.15]	−0.01	+0.15 [+0.11, +0.18]
SIM V8 INDEX	−0.47 [−0.52, −0.42]	−0.52 [−0.58, −0.46]	−0.33 [−0.39, −0.26]	−0.49	+0.17 [+0.13, +0.20]
SIM V9 GBM	+0.23 [+0.19, +0.27]	+0.24 [+0.19, +0.28]	+0.21 [+0.16, +0.27]	+0.22	+0.38 [+0.33, +0.43]
SIM V9 INDEX	+0.05 [−0.01, +0.10]	−0.01 [−0.07, +0.04]	+0.21 [+0.15, +0.27]	+0.02	+0.68 [+0.65, +0.71]
SIM V10 GBM	−0.03 [−0.07, +0.01]	+0.02 [−0.02, +0.06]	−0.17 [−0.21, −0.12]	−0.05	+0.12 [+0.10, +0.15]
SIM V10 INDEX	−0.39 [−0.44, −0.34]	−0.44 [−0.50, −0.38]	−0.25 [−0.31, −0.19]	−0.41	+0.25 [+0.21, +0.27]
SIM B corr GBM	+0.79 [+0.67, +0.91]	+0.76 [+0.60, +0.91]	+0.88 [+0.75, +0.98]	+0.65	+0.94 [+0.77, +1.11]
SIM B corr INDEX	−0.11 [−0.15, −0.08]	−0.16 [−0.20, −0.12]	+0.00 [−0.04, +0.05]	−0.13	+0.52 [+0.50, +0.54]

S7. CONTROLLED SIMULATION OF THE FIRST ANALYSIS ROUND

Table S18 lists the controlled tests of our earlier analysis round (unseen vehicles, sensors and speeds on synthetic routes), and Fig. S5 the sensor-robustness experiment summarised in Table IV.

TABLE S18
CONTROLLED GENERALISATION TESTS OF THE FIRST ANALYSIS ROUND IN THE PARAMETRIC SIMULATOR, WITH CRITERIA FIXED IN ADVANCE (UNSEEN VEHICLES: RELATIVE MAE INCREASE $\leq$10% INCL. CI; SPEED: |CHANGE| $\leq$10% INCL. CI).

test	model	result	criterion_met
unseen vehicles (random split)	M2a RMS only (alpha=0)	MAE seen 1.365 -> unseen 1.435; diff +0.070 [+0.022, +0.120]	True
unseen vehicles (random split)	M2b RMS/v^0.5	MAE seen 1.066 -> unseen 1.201; diff +0.135 [+0.097, +0.177]	False
unseen vehicles (random split)	M2c RMS/v^alpha (alpha fitted)	MAE seen 1.042 -> unseen 1.200; diff +0.158 [+0.117, +0.204]	False
unseen vehicles (random split)	M2d legacy exponent 2	MAE seen 2.139 -> unseen 2.097; diff -0.042 [-0.110, +0.025]	True
unseen vehicles (random split)	M3 LightGBM	MAE seen 0.742 -> unseen 0.884; diff +0.142 [+0.113, +0.176]	False
unseen vehicles (random split)	M3 LightGBM (no speed)	MAE seen 1.025 -> unseen 1.036; diff +0.011 [-0.026, +0.049]	True
unseen vehicles (random split)	M5 1D-CNN	MAE seen 0.809 -> unseen 1.004; diff +0.195 [+0.167, +0.224]	False
unseen vehicles (random split)	M5 1D-CNN (no speed)	MAE seen 1.137 -> unseen 1.358; diff +0.222 [+0.188, +0.256]	False
vehicle extrapolation (f_s > 1.25 Hz unseen)	M2a RMS only (alpha=0)	MAE low-f_s 1.669 vs high-f_s 1.423; diff -0.246 [-0.332, -0.161]	True
vehicle extrapolation (f_s > 1.25 Hz unseen)	M2b RMS/v^0.5	MAE low-f_s 1.431 vs high-f_s 1.018; diff -0.413 [-0.513, -0.314]	True
vehicle extrapolation (f_s > 1.25 Hz unseen)	M2c RMS/v^alpha (alpha fitted)	MAE low-f_s 1.407 vs high-f_s 0.971; diff -0.437 [-0.531, -0.345]	True
vehicle extrapolation (f_s > 1.25 Hz unseen)	M2d legacy exponent 2	MAE low-f_s 2.270 vs high-f_s 2.141; diff -0.128 [-0.219, -0.039]	True
vehicle extrapolation (f_s > 1.25 Hz unseen)	M3 LightGBM	MAE low-f_s 0.819 vs high-f_s 1.285; diff +0.466 [+0.416, +0.515]	False
vehicle extrapolation (f_s > 1.25 Hz unseen)	M3 LightGBM (no speed)	MAE low-f_s 1.082 vs high-f_s 1.487; diff +0.405 [+0.349, +0.464]	False
vehicle extrapolation (f_s > 1.25 Hz unseen)	M5 1D-CNN	MAE low-f_s 0.782 vs high-f_s 1.261; diff +0.478 [+0.377, +0.589]	False
vehicle extrapolation (f_s > 1.25 Hz unseen)	M5 1D-CNN (no speed)	MAE low-f_s 1.058 vs high-f_s 1.082; diff +0.024 [-0.038, +0.080]	True
BeamNG: vehicle held out (S2 vs S1)	M0 constant (training median)	MAE seen 7.60 -> unseen 7.60; diff +0.00 [-0.00, +0.01]	True
BeamNG: vehicle held out (S2 vs S1)	M1 speed-only (LightGBM on speed features)	MAE seen 7.49 -> unseen 7.61; diff +0.12 [+0.00, +0.23]	True
BeamNG: vehicle held out (S2 vs S1)	M2a RMS only (alpha=0)	MAE seen 6.68 -> unseen 7.27; diff +0.59 [+0.43, +0.73]	False
BeamNG: vehicle held out (S2 vs S1)	M2b RMS/v^0.5	MAE seen 6.36 -> unseen 6.88; diff +0.52 [+0.37, +0.66]	False
BeamNG: vehicle held out (S2 vs S1)	M2c RMS/v^alpha (alpha fitted on training data)	MAE seen 6.34 -> unseen 6.88; diff +0.54 [+0.40, +0.69]	False
BeamNG: vehicle held out (S2 vs S1)	M2d RMS*(v_ref/v)^2 (legacy exponent, gravity removed)	MAE seen 6.50 -> unseen 6.70; diff +0.20 [+0.11, +0.29]	True
BeamNG: vehicle held out (S2 vs S1)	M3 LightGBM (all features)	MAE seen 6.02 -> unseen 7.25; diff +1.24 [+0.96, +1.51]	False
BeamNG: vehicle held out (S2 vs S1)	M3 LightGBM (no speed features)	MAE seen 6.01 -> unseen 7.39; diff +1.37 [+1.10, +1.64]	False
BeamNG: vehicle held out (S2 vs S1)	M5 1D-CNN	MAE seen 8.63 -> unseen 10.95; diff +2.32 [+1.78, +2.88]	False
speed 40->100 km/h (sweep_iso)	M2a RMS only (alpha=0)	rel. change +63.3% [+60.8, +65.5]	False
speed 40->100 km/h (sweep_iso)	M2b RMS/v^0.5	rel. change +13.7% [+11.8, +15.4]	False
speed 40->100 km/h (sweep_iso)	M2c RMS/v^alpha (alpha fitted)	rel. change +3.1% [+1.4, +4.7]	True
speed 40->100 km/h (sweep_iso)	M2d legacy exponent 2	rel. change -52.0% [-52.5, -51.5]	False
speed 40->100 km/h (sweep_iso)	M3 LightGBM	rel. change +0.6% [-1.1, +2.5]	True
speed 40->100 km/h (sweep_iso)	M3 LightGBM (no speed)	rel. change +35.7% [+31.9, +38.4]	False
speed 40->100 km/h (sweep_iso)	M5 1D-CNN	rel. change -4.4% [-5.9, -2.5]	True
speed 40->100 km/h (sweep_iso)	M5 1D-CNN (no speed)	rel. change +22.1% [+19.9, +24.2]	False
speed 40->100 km/h (sweep_iso)	raw RMS of a_z (no model, not an IRI)	rel. change +82.5% [+79.1, +85.4]	False
speed 40->100 km/h (sweep_p79)	M2a RMS only (alpha=0)	rel. change +166.7% [+158.7, +171.4]	False
speed 40->100 km/h (sweep_p79)	M2b RMS/v^0.5	rel. change +96.1% [+89.7, +99.9]	False
speed 40->100 km/h (sweep_p79)	M2c RMS/v^alpha (alpha fitted)	rel. change +78.3% [+72.5, +81.8]	False
speed 40->100 km/h (sweep_p79)	M2d legacy exponent 2	rel. change -32.0% [-33.5, -31.1]	False
speed 40->100 km/h (sweep_p79)	M3 LightGBM	rel. change +60.2% [+50.4, +67.7]	False
speed 40->100 km/h (sweep_p79)	M3 LightGBM (no speed)	rel. change +92.2% [+62.2, +117.4]	False
speed 40->100 km/h (sweep_p79)	M5 1D-CNN	rel. change +25.2% [+22.5, +27.4]	False
speed 40->100 km/h (sweep_p79)	M5 1D-CNN (no speed)	rel. change +73.1% [+47.9, +94.2]	False
speed 40->100 km/h (sweep_p79)	raw RMS of a_z (no model, not an IRI)	rel. change +235.0% [+223.2, +242.3]	False
speed 40->100 km/h (sweep_p79_trained_on_p79 (LORO))	M2c RMS/v^alpha (alpha fitted)	rel. change +7.4% [+5.2, +10.0]	False
speed 40->100 km/h (sweep_p79_trained_on_p79 (LORO))	M3 LightGBM	rel. change -0.9% [-4.0, +2.0]	True
speed 40->100 km/h (sweep_p79_trained_on_p79 (LORO))	M3 LightGBM (no speed)	rel. change +35.6% [+16.7, +51.4]	False

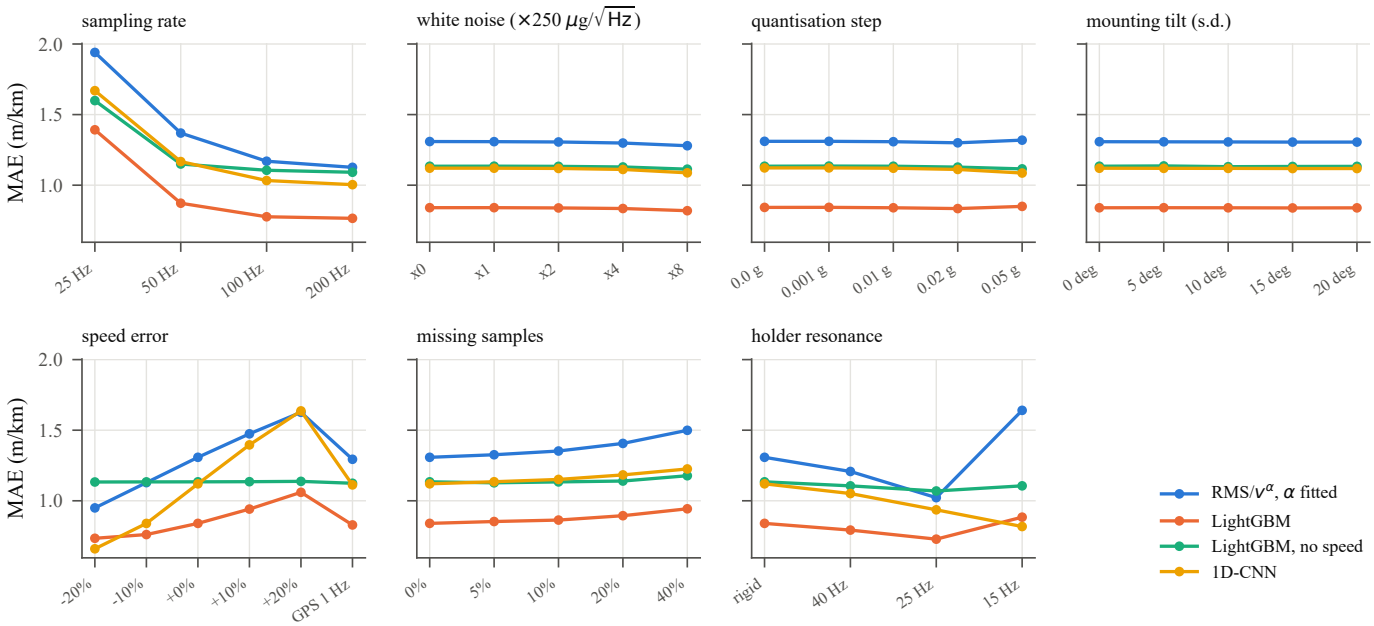


Fig. S5. Sensor robustness in simulation: estimators trained on synthetic profiles, evaluated on 12 held-out synthetic routes driven by 6 unseen vehicles with the telematics sensor configuration; one property changed at a time (common random numbers).

S8. Reproduction

Every number in the paper and this supplement is a LaTeX macro generated from the result files (`make_numbers.py`, `make_numbers_v2.py`); tables and figures are generated by `make_tables.py`, `make_tables_v2.py`, `make_figures.py` and `make_figures_v2.py`. The complete pipeline, including the second analysis round, is `run_all_experiments.py`; `README.md` lists the commands, the pinned environment (`environment_lock.txt`), seeds, runtimes and the provenance of every headline number.